Marketing Campaign Performance

Where does the ad budget go?

An investigation into 200,000 advertising campaigns



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The question: what makes ads pay off?

200,000
campaigns analysed

2,500.9 M
USD of advertising budget

8.80 M
conversions (new customers)

Our yardstick: cost per conversion

What we pay for each new customer. Lower is better.

334.4 USD
typical (median) campaign

Conversions estimated as clicks × conversion rate. Budget taken as total campaign spend. Source: Kaggle, 2021 data.

The suspects: none of the usual ones

Cost per conversion by channel (USD)



Median per campaign, bars start at 0. The largest gap is 2.1%, within the range of chance.

Who we target

Every audience, segment, city and language: 331.5 to 337.4 USD.

A winning formula

Of the 10 best of 150 combinations, 0 repeat in a second random sample.

Bigger budgets

Median conversions stay at 35.5 to 36.0 from the smallest to the largest budgets.

Differences tested with Kruskal-Wallis. Campaign length and calendar month are flat too.

The twist: a leak in every channel

9.2%

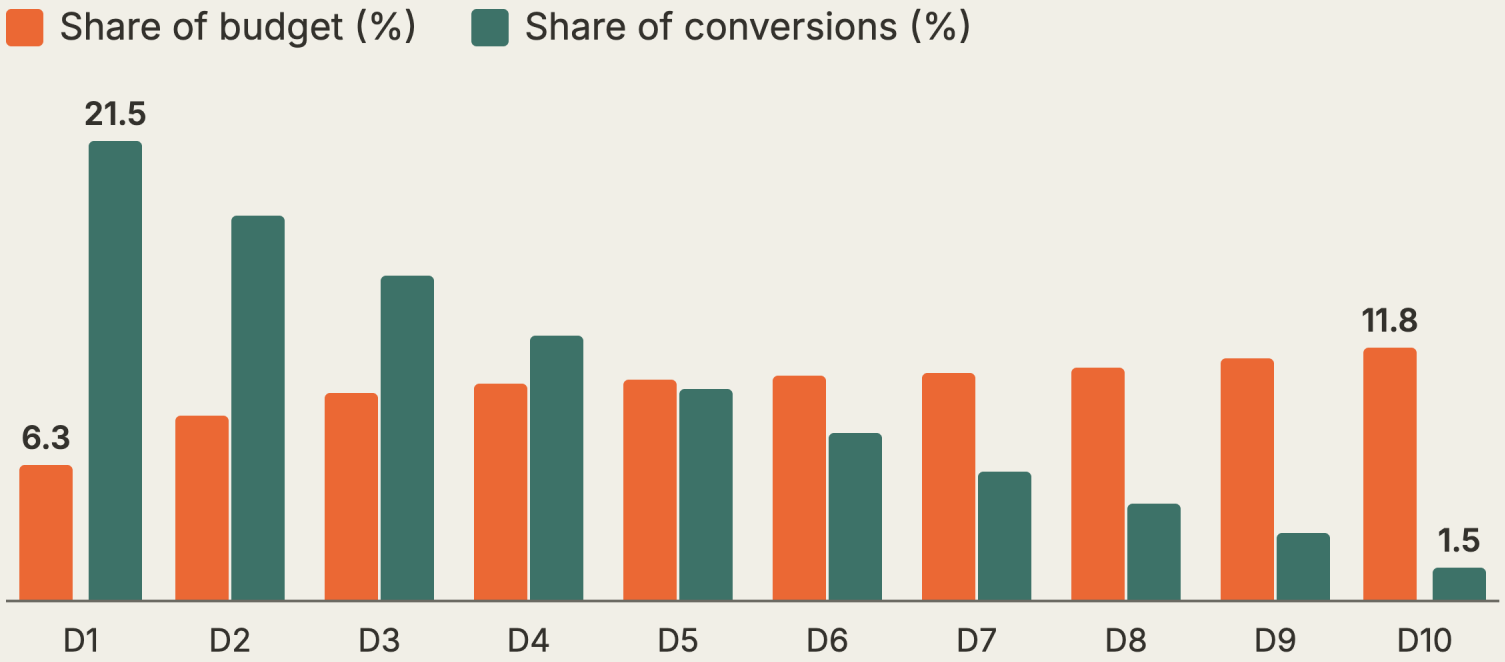
of campaigns (the high-cost ones)

10.9%

of the budget: 272.1 M USD

1.3%

of the conversions



Campaigns ranked by cost per conversion, in 10 equal groups: D1 is the cheapest 10%, D10 the most expensive.

High-cost = statistical outliers in cost per conversion (IQR rule). They are 9.1% to 9.5% of campaigns in every channel.

The fingerprint: very low conversion

One simple rule: flag every campaign whose final conversion rate is 2% or less.

10.7%
of all campaigns get flagged

58.4%
of the high-cost campaigns are caught

50.3%
of the flagged ones really are high-cost

A weaker clue: very large budgets

High-cost campaigns go from 2.9% of the smallest budgets to 16.0% of the largest, because more spend brings no more conversions.

Still to prove: how early it shows

We only have final results. Proposal: a pilot that compares first-week conversion with the final outcome.

Budget groups are quintiles of campaign budget. Setup choices do not predict high-cost campaigns: 8.9% to 9.6% in every group.

The blind spots: what the data cannot say

No retention

We see who arrives, not who stays. Engagement score is the closest proxy, and it is flat.

No daily results

Only the end result of each campaign, and dates assigned in rotation by campaign ID.

ROI does not add up

High-cost campaigns show the same median ROI as the rest: 5.00 against 5.01.

Synthetic data

Columns are unrelated to each other and values are unrealistic, such as a median CTR of 10%.

Read the method as the takeaway. The exact figures need real data to confirm them.

The plan: four moves

1 **Review campaigns that convert at 2% or less**
Half of them are high-cost. Test how early the signal appears.

2 **Put the largest budgets under closer watch**
16.0% of them are high-cost, and more spend brought no more conversions.

3 **Keep the budget split between channels and audiences**
On this data, moving money between them has no measured benefit.

4 **Measure what is missing**
Day 7 and day 30 retention, an ROI that includes costs, real dates and A/B tests.

The proof: see it live

marketing-campaigns-rita.streamlit.app

Pick a channel

The leak appears again: 9.1% to 9.5% of campaigns are high-cost in every channel.

Move the rule

At 5%, it catches 89.7% of the high-cost campaigns, but only 25.8% of those flagged are.

Thank you. Code, notebook and full report are on GitHub: IA-P1-BCN/EDA_project_by_Rita